

A 1:10 Autonomous Car for the Team

Budget breakdown under \$2,000

Formula Electric at Berkeley, Autonomous Team

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Summary

We ask for **\$1,750** to build one 1:10-scale autonomous car and a cardboard lab track, with a minimum configuration at **\$1,100** if the budget is tighter. It uses the sensors the team already owns where they fit. The car is the RoboRacer-standard Traxxas Slash with a 2D lidar, a small computer and a motor controller that reports wheel speed. It is what new members learn real robotics on, what the perception and control work is tested on between FSAE seasons, and the first place our code meets real dynamics. The team's simulator already exists and is in use; this document is about the hardware, area by area, with the costs, the choices inside each area, and the realities that come with them.

Area	Recommended	Minimum
Chassis and drive	500	330
Motor controller	110	110
Compute	250	130
Lidar	320	100
Camera	0	0
Power	230	200
Mounting and wiring	80	60
Track and cones	45	15
Spares and tools	60	40
Subtotal	1,595	985
Contingency (10 %)	160	100
Total	1,755	1,085

USD, September 2026 list prices, rounded. Verify at order time; the lidar moves most.

1 Where we are

- **Simulator.** The team has a working simulator of this exact car, a ROS 2 devkit, a starter driving stack and a leaderboard. Members already write and race code in it. It needs nothing from this budget. Its limit is that nothing in it is real: no grip change with the floor, no servo backlash, no battery sag, no lidar seeing chair legs.
- **Owned hardware.** An Intel RealSense depth camera, a RoboSense M1 3D lidar, and a motorless Tamiya 1:10 touring chassis. The camera goes on this car; the other two are discussed in section 3.
- **Space.** A lab corridor is enough for a 1:10 track. The Alameda test track is for larger vehicles and is not needed here.

2 The car, area by area

2.1 Chassis and drive

Option	USD	
Traxxas Slash 4x4 VXL (TRA68086-4), brushless motor and ESC included	500	recommended
Traxxas Slash 2WD VXL (TRA58076-4), brushless	330	minimum

The Slash is the chassis every RoboRacer build and drawing assumes, and the one the simulator models. The 4x4 is the standard: it carries the sensor deck without lifting the front wheels and tolerates novice driving. The 2WD saves \$170 and is fine for lidar-only driving on a flat floor, but it spins the rear under a heavy deck and is not what the drawings fit; buy it only if the total must come under \$1,200. Either comes ready to run with a brushless motor; the stock speed controller is replaced by the item below, and the radio and receiver stay as the manual-override link.

2.2 Motor controller

Option	USD	
Flipsky mini FSESC 6.7 (VESC-compatible)	110	recommended
VESC 6 MkVI (reference part)	260	not needed at this budget

A VESC-type controller is not optional: it is what gives the software wheel speed (odometry), current limits, and a clean serial interface. The reference VESC costs \$150 more for better cooling and support; the Flipsky runs the same firmware and is what most student builds use.

2.3 Compute

Option	USD	
NVIDIA Jetson Orin Nano developer kit	250	recommended
Raspberry Pi 5, 8 GB, active cooler, SD card, PSU	130	minimum

The Pi runs ROS 2 and a lidar-only driver comfortably and is the cheaper way to a first lap. It cannot run camera perception at useful rates, so the day the team wants cone detection from the RealSense, it needs the Jetson. Buying the Jetson first avoids buying twice; buying the Pi first gets a car on the floor for \$120 less and keeps the Jetson as the upgrade.

2.4 Lidar

Option	USD	
Slamtec RPLidar A2M12, 12 m, 12 Hz, 0.25°	320	recommended
Slamtec RPLidar A1M8, 12 m, 5 to 10 Hz, 1°	100	minimum
Hokuyo UST-10LX, 10 m, 40 Hz, 0.25°	1,500	the reference; out of this budget

The lidar decides how fast the car can safely go. The reference Hokuyo scans at 40 Hz and is what the simulator models; it is also most of a \$3,000 build. The A2M12 at 12 Hz is a sensible car for 2 to 3 m/s and is what the budget allows. The A1M8 is a \$100 sensor that proves the pipeline and nothing more: 8 kHz sample rate, a degree of angular resolution, noisy beyond 6 m. Spinning lidars are also the part that breaks in a crash; mount it behind the bumper line.

2.5 Camera

The team's RealSense goes on the car at no cost. With the lidar it gives the sensor pair for cone detection: the camera classifies cones by colour, the lidar gives range and bearing. This is the perception work that carries most directly to the FSAE car, and it needs the Jetson.

2.6 Power

Item	USD	
Two Traxxas 3S 5000 mAh LiPo packs	170	one drives, one charges
Balance charger	50	
LiPo safety bag	10	
5 V / 5 A buck converter, XT60 splitter, inline fuse	30	feeds computer and lidar from the drive pack

Minimum: one pack (\$85 less). The RoboRacer powerboard (\$100) is a cleaner version of the converter and splitter with a safe-shutdown line; not needed to start.

2.7 Mounting and wiring

Laser-cut deck plates from the published RoboRacer drawings, cut in a campus makerspace (about \$30 of acrylic), standoffs and fasteners (\$15), a powered USB hub and cables (\$35). Minimum \$60 with 3D-printed brackets instead of a cut deck.

2.8 Track and cones

Item	USD	
Cardboard walls: flattened boxes cut into 20 cm strips, taped in loops	0	
Floor tape (start line, lane marks)	15	
FSAE-style cones, 3D-printed at 1:10, 100 pieces	15	about 10 g of filament each
Foam pipe insulation on the corners	15	

Cardboard is enough for the lab: the lidar sees it as well as any wall, a 20 cm strip is well above the scan plane, and a wall that folds when hit is kinder to the car. It has to be re-taped each session and does not store. The cones are the part worth printing on day one, because the cone track is where the camera work happens. *Later, if the track is in daily use, 30 m of 33 cm flexible aluminium duct (about \$130) is the permanent version.*

2.9 Spares and tools

Tyres (\$30), pinion and bumper (\$20), zip ties, tape, Velcro (\$10). The team's own tools otherwise. A wall hit at 3 m/s breaks a bumper or a servo saver, not the car; a lidar hit breaks the lidar.

3 Options considered and set aside

- **The team's Tamiya chassis.** A motorless touring-car kit. It would save the \$500 chassis line and cost a motor, controller and servo (\$200) plus a custom sensor deck with no drawings to copy. Its wheelbase, track width and ground clearance differ from the car the simulator models and from every published build, so nothing transfers and every mounting problem

is solved alone. Net saving about \$300 for a semester of mechanical work that teaches no autonomy. It stays a bench mule for wiring and sensor practice.

- **A 1:4 or 1:5 car with the team's M1 lidar.** No maker sells a 1:4 road car; the nearest are the large 8S trucks: the Traxxas X-Maxx, the Arrma Kraton 8S and the Losi 5IVE-T. With the owned M1 and the Alameda track counted, a 1:5 build costs about \$3,300 to \$3,900: chassis \$1,300 to \$1,900, 8S batteries and charger \$500, a high-current controller \$350, a Jetson Orin NX class computer for the M1's point clouds \$700, deck, roll protection and kill switch \$450. Twice this budget, and it cannot be run on campus: a 10 kg car at 50 km/h needs a closed outdoor area, so every test is a session in Alameda rather than an afternoon in the lab, and it is not a car new members can learn on. The M1 itself (about 0.7 kg, a half-metre blind zone, 120° forward view) is a full-size-vehicle sensor that does not fit a 1:10 car's mass or corridor. Both wait for a later budget, after the 1:10 programme has run.

4 What the money buys, in order

When	Milestone
Order + 2 weekends	Car assembled from the RoboRacer build guide, radio drive, all sensors on ROS 2
+ 3 weeks	Starter stack drives the cardboard track; simulator-to-car differences written down
+ 6 weeks	Members' own stacks on the car; internal time-attack; first cone-detection results from the camera
Semester end	Perception and control lessons carried into the FSAE autonomous work

5 The realities

- **One car.** Access has to be scheduled and earned; the simulator's qualification rule is the gate. Expect queueing in the weeks before an internal event.
- **The cheap parts cap performance.** At 12 Hz the lidar and the Pi limit safe speed to about 2 to 3 m/s; the simulator car goes faster. Members will feel the gap; that is the point, but it should be said up front.
- **Time is the scarce resource, not money.** Two weekends of assembly, a month to a first clean lap, and someone has to own the car, the batteries and the charging procedure. If nobody owns it, it becomes a shelf ornament by week six.
- **Batteries need a procedure.** LiPo bag, supervised charging, no charging unattended, a written checklist before the first charge.
- **The floor matters.** Lab tile and carpet grip differently; the car will be tuned for one floor. A cardboard track moves between sessions; lap times will not be comparable week to week until the duct arrives.
- **Cardboard walls have a life of weeks.** Budget half an hour per session for the track.
- **Out of scope for this budget.** The 1:5 car with the M1, a second car, and physical competitions. All are possible later; none are needed to get value from this car.

RoboRacer build guide and drawings: <https://f1tenth.readthedocs.io>

Team platform (simulator, devkit, manuals): <https://github.com/Pranman1/feb-racing>